Marcus Chan

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Experience

Custom Software Engineering Associate Manager
2021 – Present

Accenture — *Singapore*

- **Technical Lead** - Aviation (2024 - Present)
  - Led the architecture and delivery of enterprise-scale serverless and generative AI applications on AWS (Lambda, AppSync, CDK, DynamoDB, Amplify, Bedrock, SageMaker) at client sites
  - Drove technical strategy and engineering decisions, translating aviation business requirements into scalable production systems
  - Designed and delivered full-stack solutions including an LLM-based document classification pipeline, RFID showcase systems, mobile wayfinding applications, and e-commerce platforms
  - Served as primary technical liaison between engineering and business teams, aligning delivery with operational requirements and passenger experience goals
  - Spearheaded infrastructure automation, CI/CD pipeline implementation, and knowledge transfer programmes to enable client self-sufficiency
  - Mentored junior engineers and conducted code reviews to uphold engineering standards across the team
- **Site Reliability Engineer** - Public Services (2022 - 2024)
  - Designed and deployed an enterprise chatbot solution that reduced service request processing time by 65–70%
  - Implemented monitoring and alerting frameworks across multi-tier applications and infrastructure
  - Streamlined tenant onboarding processes for secure and repeatable platform integration
  - Hardened system security through proactive vulnerability remediation and configuration governance
- **Software Development Engineer** - Financial Services (2022)
  - Built backend Django REST APIs for an enterprise financial services platform, partnering directly with SMEs to translate complex financial calculation logic into reliable, maintainable code
  - Represented Accenture's engineering capabilities at Singapore FinTech Festival 2022, demonstrating the platform to senior client stakeholders and industry partners
- **Application Engineer** - Healthcare Services (2021 - 2022)
  - Automated complex reporting workflows through data analysis and SSRS/SSIS pipeline development
  - Resolved critical service requests across SSRS/SSIS and Pega systems, maintaining 99.9% system uptime
  - Delivered operational support to healthcare stakeholders, ensuring continuous service availability

Software Delivery Engineer
2019 – 2021

Whale Cloud — *Singapore*

- **Software Delivery Engineer** - (2020 - 2021)
  - Delivered enterprise-grade Business Intelligence dashboards using Alibaba Cloud ecosystem (Dataphin, DataWorks, QuickBI, DataV) across multiple client engagements
  - Recognised by Alibaba Cloud as a subject matter expert; invited to conduct technical sharing sessions on DataV platform capabilities
  - Led end-to-end UAT cycles and QA processes, achieving zero post-release defects across client deliverables
  - Managed POC development initiatives as technical product manager, accelerating time-to-market for new solutions
- **Software Engineer** - (2019 - 2020)
  - Joined Whale Cloud Singapore through the IMDA Company-Led Training Programme and converted to permanent staff within 12 months
  - Built and supported big data implementations including ETL pipeline development, advanced SQL analytics, and BI dashboard configuration
  - Contributed to Alibaba Cloud DataWorks and Dataphin deployments, supporting data engineering and UAT activities across client projects

Junior Software Developer
2017 – 2018

Veev Interactive — *Singapore*

- Developed interactive e-learning content using Articulate 360, JavaScript, and Adobe Creative Suite (Premiere Pro, After Effects, Illustrator, Audition)
- Improved content accessibility by implementing audio transcription and multi-language subtitles across all courseware
- Streamlined production workflows through systematic content reviews, version control, and asset management processes

Public Transport Security Command Officer
2014 – 2016

Singapore Police Force — *Singapore*

- National Service: mandatory two-year military enlistment for Singapore male citizens.
- Led patrol teams across public transportation networks to maintain security and public safety
- Maintained operational readiness and rapid response capabilities for security incidents

IT System Architecture Research Intern
2013 – 2013

Hitachi — *Singapore & Yokohama, Japan*

- **Research & Development Center - Singapore**
  - Built a Taxi Simulator application using HTML, Java, JavaScript, and Google Maps API to model real-world transportation patterns
  - Integrated Open Source Routing Machine (OSRM) to simulate taxi interactions and optimise passenger demand forecasting
  - Created data visualisation dashboards analysing taxi sharing efficiency and route optimisation using historical traffic data
- **Yokohama Research Lab - Yokohama, Japan**
  - Developed a scalable Big Data Graph Visualiser web application using Apache Solr and Apache Hadoop for large-scale data processing
  - Built Java backend services and JavaScript frontend components enabling interactive exploration of complex graph datasets
  - Conducted performance benchmarking of graph algorithms on distributed Linux systems, identifying optimal configurations for production use

Technical Skills

Languages: Java, Python, SQL, TypeScript
Fullstack: Django, FastAPI, GraphQL, Next.js, Vite / PWA
Cloud: AWS CDK, AWS Serverless (Lambda, AppSync, DynamoDB), Alibaba Cloud Data & BI (DataWorks, Dataphin, QuickBI, DataV), Amplify, Event-driven & Observability (SNS, CloudWatch)
Platform: CI/CD & Automation, Databricks, Docker, Terraform
Ai: Amazon Bedrock, Enterprise Chatbots, LLM Document Classification, Local LLMs (Ollama), SageMaker
Leadership: Agile / Scrum (CSM, CSPO), Jira, Mentoring & Code Review, Technical Leadership

Certifications

AWS Certified Developer – Associate - Amazon Web Services 2026
AWS Certified AI Practitioner - Amazon Web Services 2025
HashiCorp Certified: Terraform Associate (003) - HashiCorp 2024
Databricks Certified Data Engineer Associate - Databricks 2024
Cloud Digital Leader - Google Cloud 2023
Certified Scrum Product Owner® (CSPO®) - Scrum Alliance 2022
AWS Certified Cloud Practitioner - Amazon Web Services 2022
Certified System Architect (CSA) - Pegasystems 2021
ITIL® Foundation Certificate in IT Service Management - Axelos 2021
Certified ScrumMaster® (CSM®) - Scrum Alliance 2021
Alibaba Cloud Professional - Cloud Computing - Alibaba Cloud 2019
Alibaba Cloud Associate - Cloud Computing - Alibaba Cloud 2019

Education

The University of Queensland — Brisbane, Australia
Bachelor of Information Technology with a Major in Software Information Systems — *2018 – 2018*

Singapore Management University — Singapore
Bachelor of Science (Information Systems) (Incomplete) — *2016 – 2017*

Temasek Polytechnic — Singapore
Diploma in Information Technology — *2011 – 2014*